

Multi-Market Stock Volatility Forecasting Model

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Abstract. Forecasting daily stock-price volatility, how much a stock’s price is likely to swing on a given day, is a core input to risk management and option pricing: a risk manager sizing exposure or a trader pricing an option needs to know not only which direction a price may move but how large its swings will be. We propose a per-stock volatility forecasting model built on a gamma-loss gradient-boosting machine (GBM) over own-history features, and we ask where extra information actually helps. On the S&P 500, adding an External, forward-looking scheduled-earnings feature (GBME) cuts QLIKE by 8.7 to 11.2% over own-history at every horizon (Diebold-Mariano $p < 0.001$), roughly seven times the size of any graph term. A cross-firm graph feature, by contrast, adds nothing once own-history and the firm’s own momentum are in the model: it is not distinguishable from the earnings model at the short horizons and it hurts at the monthly horizon. We replicate the study on a second, structurally different market, 405 Vietnam Ho Chi Minh Stock Exchange (HOSE) tickers, under an identical leakage-controlled walk-forward protocol scored on QLIKE with date-clustered Diebold-Mariano tests. The graph null replicates on HOSE, while the earnings lever does not transfer, which localizes the earnings gain to the S&P 500 and shows that the graph carries no reliable signal on either market.

Keywords: Volatility forecasting · HAR · Gradient boosting · Earnings announcements · Diebold-Mariano.

1 Introduction

Financial markets are volatile: prices swing, and the size of those swings, an asset’s *volatility*, is the standard measure of its risk. Because volatility drives how options are priced, how much capital a position ties up, and how risk is budgeted across a portfolio, forecasting it accurately is a long-standing problem in financial econometrics [1]. Volatility is not observed directly; it is estimated from prices, so any forecast is a forecast of an estimate. We target the daily *Parkinson estimator* [2], which gauges a day’s variance from its high-low price range and uses more of the day’s information than a close-to-close measure; our forecasting target is therefore a variance (a squared quantity), not a standard deviation.

The standard linear approach is the HAR family [1]: a plain linear regression that predicts tomorrow’s volatility from its own averages over the last day,

the last week (about five trading days), and the last month (about twenty-two), summarising volatility’s long memory with a handful of coefficients. This benchmark is deliberately simple yet hard to beat. A recent literature argues that HAR treats each stock in isolation and misses cross-firm spillover, and wires stocks together with graph neural networks that aggregate a stock’s correlated neighbors, reporting gains on squared-error metrics. We test whether such cross-firm structure helps once the comparison is fair, and we contrast it with a different kind of information: a forward-looking, per-firm signal.

We forecast daily Parkinson variance for the S&P 500 with a per-stock gamma-loss gradient-boosting model over own-history features, then add two candidate levers and measure each one under a leakage-controlled walk-forward protocol scored on QLIKE with date-clustered Diebold-Mariano tests. The lever that works is forward-looking and specific to the firm. A scheduled earnings release is known weeks ahead and raises volatility around the announcement [9]; encoding the distance from the forecast target to the nearest scheduled release cuts QLIKE by 8.7 to 11.2% over own-history at every horizon, roughly seven times the size of any graph term. A cross-firm graph feature adds no reliable value: the short-horizon signal it could carry is redundant with the stock’s own momentum, and its remaining signal arrives contemporaneously with the target rather than at the forecast origin.

This paper contributes three results. First, a per-stock gamma-GBM over own-history features is a strong, transparent baseline for daily volatility that already beats HAR and HARQ. Second, a forward-looking scheduled-earnings feature is the dominant lever on the S&P 500, cutting QLIKE by 8.7 to 11.2% at every horizon (Diebold-Mariano $p < 0.001$), while a cross-firm graph feature adds nothing. Third, we replicate the study on a second market, 405 Vietnam HOSE tickers, where the graph null holds and the earnings lever does not transfer, which localizes the earnings gain and shows the graph carries no reliable signal on either market.

Terminology. *Volatility*: the size of a stock’s price fluctuations, the standard proxy for its risk. *Parkinson estimator*: a day’s variance measured from its high-low price range; our forecasting target is this variance (squared units), not the standard deviation. *HAR*: a linear model predicting volatility from its own daily, weekly, and monthly averages. *QLIKE*: a proxy-robust error measure for volatility forecasts that stays reliable when the target is a noisy proxy [4]. *Diebold-Mariano (DM) test*: a test of whether one forecast is more accurate than another by more than chance, computed from their day-by-day loss differences [5]. *Horizon h* : how many trading days ahead we forecast ($h \in \{1, 5, 10, 22\}$, roughly a day, week, fortnight, and month).

2 Method and Setup

Target and metric. We forecast daily Parkinson variance $\text{pk}_t = \ln(H_t/L_t)^2/(4 \ln 2)$ from the daily high H and low L ($\ln$ is the natural logarithm), at horizons

$h \in \{1, 5, 10, 22\}$ over 497 S&P 500 constituents. We report QLIKE as the primary loss [4], alongside MSE, RMSE and MAE. We test significance with a Diebold-Mariano statistic [5] on per-trading-date mean loss, with the Harvey-Leybourne-Newbold small-sample correction [6], because within-day stock errors are dependent.

Model. The forecasting model is a per-stock GBM (a gamma-loss Gradient Boosting Machine [7]: scikit-learn HistGradientBoostingRegressor [8], gamma-deviance loss equal to QLIKE up to a constant, 300 iterations, learning rate 0.05, 31 leaves, L2=1.0, 3-seed ensemble). Its own-history feature vector is $\mathbb{R}^9$: three HAR lags [1], a realized-quarticity proxy, and five log-volatility momentum terms. Two optional feature blocks extend it. The External block (GBME) is $\mathbb{R}^4$ of forward-looking scheduled-earnings features, chiefly the distance from the forecast target to the nearest scheduled release. The graph block (GBME+graph) is a single cross-firm neighbor aggregate, $\mathbb{R}^1$, formed by averaging each node’s top-10 train-correlation neighbors with positive row-normalized weights. Figure 2 shows the architecture.

Baselines. HAR [1] fits three own-history lags by ordinary least squares. HARQ adds a realized-quarticity interaction [3]. Both share the same target, protocol, and positivity floor as the GBM.

Features

Own-history block (9 features, used by GBM). Every own-history feature is observable at the forecast origin t . The three HAR lags capture short, medium, and long memory of realized variance over one day, one trading week (WK = 5), and one trading month (22 days):

$$\text{har_daily} = \text{pk}_t, \quad \text{har_weekly} = \frac{1}{5} \sum_{k=0}^4 \text{pk}_{t-k}, \quad \text{har_monthly} = \frac{1}{22} \sum_{k=0}^{21} \text{pk}_{t-k}. \quad (1)$$

A realized-quarticity proxy gauges the magnitude and dispersion of recent variance, which lets the model down-weight noisy high-variance periods:

$$\text{rq}_t = \sqrt{\frac{1}{5} \sum_{k=0}^4 \text{pk}_{t-k}^2}. \quad (2)$$

Five log-volatility momentum terms, built on $\ell_t = \ln(\max(\text{pk}_t, \varepsilon))$ with $\varepsilon = 10^{-8}$, encode recent trend, multi-scale slope, deviation from the recent mean, and a standardized 22-day level:

$$\text{mr_change} = \ell_t - \ell_{t-1}, \quad \text{mr_slope5} = \frac{\ell_t - \ell_{t-5}}{5}, \quad \text{mr_slope10} = \frac{\ell_t - \ell_{t-10}}{10}, \quad (3)$$

$$\text{mr_dev5} = \ell_t - \frac{1}{5} \sum_{k=0}^4 \ell_{t-k}, \quad \text{mr_z22} = \frac{\ell_t - \mu_{22}(\ell)}{\sigma_{22}(\ell) + \varepsilon}, \quad (4)$$

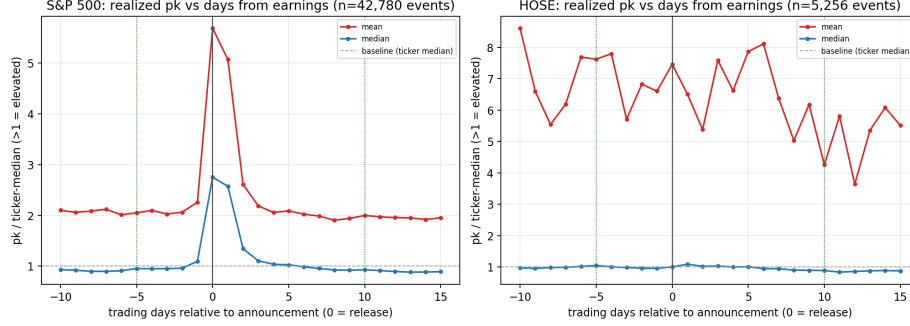


Fig. 1. Event study: median realized Parkinson variance (normalised by each ticker’s median) versus trading days from a scheduled earnings release. The S&P 500 median rises to about 2.75 times baseline on the announcement day and returns within roughly a week; the HOSE median stays near one at every offset, showing no announcement spike, which is why the earnings feature helps the S&P 500 but not HOSE.

where μ_{22} and σ_{22} are the trailing 22-day mean and standard deviation of ℓ . Together these summarize the stock’s own short-run acceleration and regime level.

External earnings block (4 features, used by GBME). Each earnings feature is evaluated at the target day $t + h$ from scheduled release dates known weeks ahead, so the block is forward-looking and leakage-safe. Let nxt be the trading days from $t + h$ to the next scheduled announcement, prv the trading days since the previous one, and $\text{dist} = \min(\text{nxt}, \text{prv})$:

$$\text{earn_prox} = \max\left(0, 1 - \frac{\text{dist}}{5}\right), \quad \text{earn_soon} = \mathbf{1}[\text{dist} \leq 3], \quad (5)$$

$$\text{earn_pre} = \max\left(0, 1 - \frac{\text{nxt}}{5}\right), \quad \text{earn_post} = \max\left(0, 1 - \frac{\text{prv}}{10}\right). \quad (6)$$

Here earn_prox is a symmetric triangular proximity that peaks at 1 on the announcement day and reaches 0 once the nearest release is at least a week away, earn_soon is a sharp ± 3 -day flag, earn_pre rises into the next release, and earn_post decays over a longer 10-day window because post-earnings volatility drift persists longer than the pre-announcement run-up. Away from any release all four are zero, so GBME reduces to GBM on ordinary days and activates only in the roughly two-week window around a scheduled release. Figure 1 shows the S&P 500 median variance rises to about 2.75 times baseline on the announcement day, which motivates these windows; on HOSE the median stays flat, so the block is inert there.

Cross-firm graph block (1 feature, used by GBM+graph and GBME+graph). The graph feature aggregates neighbours’ contemporaneous variance at the forecast origin,

$$g_{i,t} = \sum_j W_{ij} \text{pk}_{j,t}, \quad (7)$$

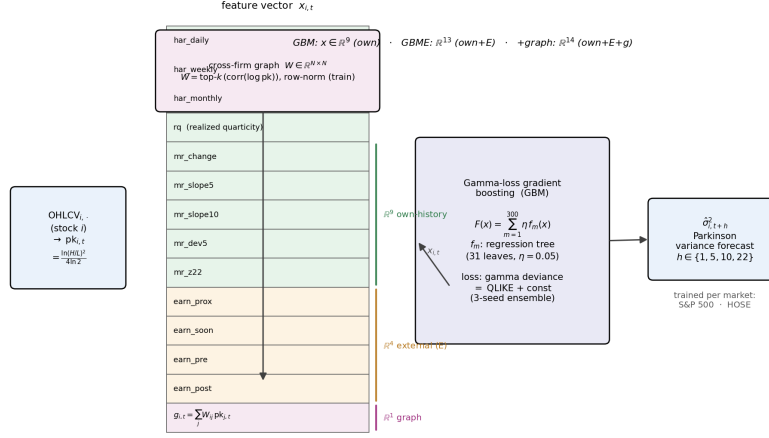


Fig. 2. Model architecture. A per-stock feature vector combines own-history features ($\mathbb{R}^9$: three HAR lags, a realized-quarticity proxy, five log-volatility momentum terms), an External forward-looking scheduled-earnings block ($\mathbb{R}^4$), and one cross-firm graph neighbor-aggregate feature ($\mathbb{R}^1$). A gamma-loss gradient-boosting ensemble maps the feature vector to the multi-horizon Parkinson-variance forecast.

where W is the row-normalized adjacency that keeps each stock’s top- $k=10$ positive train-correlation neighbours, with correlations computed from log-pk on the training window only. The aggregate is leakage-safe, but because the signal is contemporaneous it adds no incremental QLIKE beyond own-history, as the results show.

Protocol. We use expanding-window walk-forward folds with a target-horizon embargo. All correlations, adjacencies, factor estimates, and model fits use training data only; the test window is read once. Every gradient-boosting model averages three seeds. Predictions are floored identically across models before scoring.

3 Cross-Firm Graphs Add No Incremental Value

Table 1 reports all four metrics for the S&P 500 across HAR, HARQ, the own-history GBM, the earnings model GBME, and GBME+graph at every horizon. The own-history GBM already beats HAR and HARQ on QLIKE at every horizon. Adding the cross-firm graph feature to the earnings model does not help: GBME+graph is not distinguishable from GBME on QLIKE at the short horizons and it is worse at the monthly horizon. The graph is therefore never the DM-verified best, and it is not bolded in any horizon block.

The short-horizon graph signal is redundant with own-history momentum. A short-horizon graph signal appears only when the model’s own-history is stripped to three HAR lags; restoring the five log-volatility momentum features to the base erases it. The stock’s own momentum features carry what

Table 1. Main result, S&P 500 (497 constituents; walk-forward, 3-seed GBM ensemble). Lower is better on every metric. MSE $\times 10^{-7}$, RMSE/MAE $\times 10^{-4}$; QLIKE unscaled. Bold marks the best value in each metric column per horizon; the QLIKE bold additionally requires a significant date-clustered Diebold-Mariano test ($p < 0.05$).

h	Model	MSE	RMSE	MAE	QLIKE
1	HAR	5.754	7.586	2.132	0.3697
1	HARQ	5.648	7.515	2.151	0.3780
1	GBM	5.557	7.454	2.089	0.3570
1	GBME	5.397	7.346	2.017	0.3172
1	GBME+graph	5.424	7.365	2.010	0.3158
5	HAR	6.141	7.836	2.297	0.4364
5	HARQ	6.132	7.831	2.309	0.4377
5	GBM	6.117	7.821	2.255	0.4116
5	GBME	5.972	7.728	2.180	0.3705
5	GBME+graph	6.063	7.787	2.196	0.3715
10	HAR	6.214	7.883	2.356	0.4615
10	HARQ	6.205	7.877	2.358	0.4614
10	GBM	6.262	7.913	2.312	0.4307
10	GBME	6.118	7.822	2.239	0.3912
10	GBME+graph	6.240	7.899	2.272	0.3939
22	HAR	6.327	7.954	2.419	0.4855
22	HARQ	6.311	7.944	2.415	0.4831
22	GBM	6.233	7.895	2.359	0.4464
22	GBME	6.098	7.809	2.277	0.4074
22	GBME+graph	6.087	7.802	2.300	0.4110

the neighbor aggregate would otherwise supply, so the cross-sectional neighbor signal is redundant with the stock’s own past.

An oracle bound locates the missing signal in time. Replacing the neighbor value at the origin t with the neighbor value at the target day $t + h$, an illegal upper bound contemporaneous with the label, cuts QLIKE by 19.2 to 27.2% over the graph feature ($p < 0.001$ at every horizon), where the legal graph cuts it by under 1%. The cross-sectional signal exists and is large, but it arrives contemporaneously with the target, not at the forecast origin, so no origin-time graph can capture it.

Takeaway. Conditional on own-history, cross-firm graph structure adds no reliable incremental QLIKE value at any horizon on the S&P 500. The signal a graph could carry is either redundant with the stock’s own momentum or contemporaneous with the target.

4 Forward-Looking Earnings Are the Lever

A diagnostic locates the missing lever. The own-history GBM beats HAR across the calm deciles, yet its remaining QLIKE concentrates in the storm

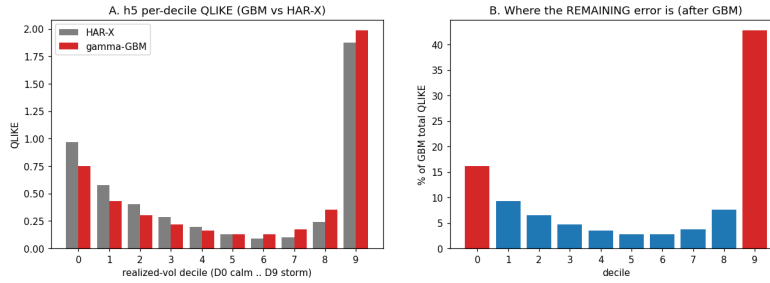


Fig. 3. The own-history GBM beats HAR on the calm deciles, and its remaining QLIKE concentrates in the storm decile D9, which carries 43% of the model total.

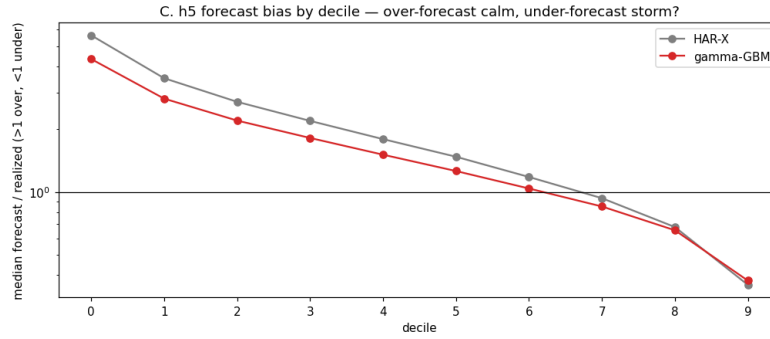


Fig. 4. The GBM under-forecasts the storm deciles, where its median forecast-to-realized ratio drops below one.

decile: decile D9 alone carries 43% of the model’s total test QLIKE (Fig. 3). The model under-forecasts exactly there, where its median forecast-to-realized ratio falls below one while every calm decile sits above one (Fig. 4). Permutation importance ranks the three own-history HAR lags far above every momentum and auxiliary term (Fig. 5), so own-history is exhausted and the storm component needs a signal from outside the stock’s past. A scheduled earnings release is known weeks ahead and drives the announcement-driven part of these spikes, which makes a forward-looking earnings feature the natural lever for the residual the GBM cannot reach.

The earnings feature is the lever. In Table 1, the External earnings model GBME posts the lowest RMSE, MAE and QLIKE at every horizon. The QLIKE gain over own-history is 8.7 to 11.2% and is significant at every horizon (DM GBME vs GBM $p < 0.001$ at $h1$, $h5$, $h10$, $h22$), which is why GBME carries the bolded QLIKE in every block. Against the HAR baseline the earnings model gains 14 to 16%. Nonlinearity and the QLIKE-loss alignment, isolated by the own-history GBM over HAR, contribute a further 3.4%. A graph added on top

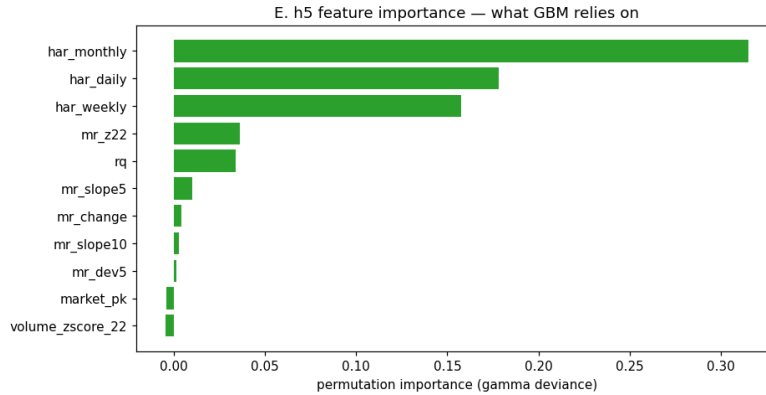


Fig. 5. Permutation importance ranks the own-history HAR lags far above every other feature, so own-history is exhausted.

of earnings changes nothing at short horizons and hurts at the monthly horizon, so the two signals do not combine.

Earnings do not fix the storm tail. Decomposed by realized-volatility decile, the model’s error still concentrates in the top decile, which carries roughly ten times the calm-day QLIKE, and its forecast-to-realized ratio there is about 0.4. The model compresses its predictions toward the conditional mean because the storm magnitude is absent from origin-time features. The residual is exogenous, which is why no cross-firm graph recovers it.

5 Related Work

Graph volatility forecasting. A recent literature wires stocks or markets through correlation or spillover edges and reports squared-error gains from graph neural networks. We test the claim on daily per-firm QLIKE with a leakage-controlled walk-forward protocol and show the gains do not survive: a cross-firm graph adds no reliable incremental QLIKE value once own-history is conditioned on. Prior Vietnamese work builds correlation and centrality networks to predict index volatility at a monthly horizon without a HAR or QLIKE benchmark; we extend that setting to daily per-firm QLIKE on 405 HOSE tickers.

HAR and its extensions. HAR [1] and HARQ [3] model volatility from own history and a realized-quarticity interaction. We keep HAR and HARQ as baselines and show that a QLIKE-loss gradient-boosting model [7, 8] with an External earnings feature dominates them without a graph, while QLIKE [4] and the Diebold-Mariano test [5, 6] provide the proxy-robust evaluation.

Table 2. Multi-market replication, HOSE (405 tickers; 390,577 to 399,040 stock-day observations per horizon). Lower is better on every metric. $\text{MSE} \times 10^{-7}$, $\text{RMSE}/\text{MAE} \times 10^{-4}$; QLIKE unscaled. Bold marks the best value in each metric column per horizon; the QLIKE bold additionally requires a significant date-clustered Diebold-Mariano test ($p < 0.05$).

h	Model	MSE	RMSE	MAE	QLIKE
1	HAR	6.604	8.126	5.193	1.8117
1	HARQ	6.389	7.993	5.062	1.7854
1	GBM	5.409	7.355	3.941	1.5702
1	GBME	5.455	7.386	3.955	1.5690
1	GBME+graph	5.560	7.456	4.000	1.5970
5	HAR	6.626	8.140	5.227	1.8218
5	HARQ	6.519	8.074	5.152	1.8075
5	GBM	5.795	7.612	4.270	1.6484
5	GBME	5.797	7.614	4.279	1.6489
5	GBME+graph	5.821	7.630	4.253	1.6487
10	HAR	6.660	8.161	5.250	1.8277
10	HARQ	6.589	8.118	5.198	1.8180
10	GBM	5.956	7.718	4.416	1.6840
10	GBME	5.960	7.720	4.429	1.6835
10	GBME+graph	5.953	7.715	4.404	1.6842
22	HAR	6.716	8.195	5.282	1.8355
22	HARQ	6.677	8.171	5.250	1.8299
22	GBM	6.213	7.882	4.615	1.7263
22	GBME	6.232	7.894	4.617	1.7261
22	GBME+graph	6.217	7.885	4.595	1.7261

6 Multi-Market Replication: Vietnam HOSE

We replicate the study on a second, structurally different market. On 405 Ho Chi Minh Stock Exchange (HOSE) tickers under the identical protocol, forecasting daily Parkinson variance at the same four horizons with 390,577 to 399,040 stock-day observations per horizon, the graph null replicates and the earnings lever does not transfer.

The own-history GBM wins and the graph adds nothing. Table 2 reports all four metrics. The own-history GBM is the DM-verified best model at every horizon: no feature block significantly lowers its QLIKE, while HAR and HARQ trail by a wide margin. Adding the graph feature ties the GBM, and the External earnings feature does not improve it: GBME is statistically indistinguishable from GBM at every horizon (DM GBME vs GBM $p = 0.79, 0.37, 0.79, 0.88$ at $h1, h5, h10, h22$). The own-history GBM therefore carries the bolded QLIKE in every block.

Earnings does not transfer to Vietnam. Using real crawled Vietnam announcement dates, GBME is statistically indistinguishable from GBM at ev-

ery horizon. A per-stock, per-day diagnostic on the HOSE test set locates two compounding causes. First, the feature fires on only 5.6% of test observations, and 0.84% before 2025, so it cannot move the pooled metric across most of the panel. Second, where it fires the volatility response is muted: realized variance on announcement-proximate days runs about 1.1 times the baseline, against the pronounced S&P 500 earnings spike, because Vietnam’s daily price limits cap the informational jump. The no-transfer result holds in the well-covered 2025-2026 folds, so it does not rest on missing dates in that window.

The oracle bound confirms the signal is contemporaneous. Replacing the neighbor value at the origin with its value at the target day cuts QLIKE by 2.6 to 3.1% over the graph feature at $h1$, $h5$ and $h10$ (DM $p = 0.004$, $p < 0.001$, $p = 0.002$), while at the monthly horizon the origin-time and target-time neighbor values are indistinguishable ($p = 0.30$). The gain is smaller than the S&P 500’s 19 to 27% and points the same way: the cross-sectional signal is contemporaneous with the label, not available at the forecast origin, so no origin-time graph captures it.

The graph null is robust to richer fusion. Two further cross-firm designs fail the same test on HOSE. Concatenating a seven-feature neighbor-spillover block into the earnings model does not beat GBME at any horizon: it is significantly worse at $h1$, $h5$ and $h22$ (DM $p < 10^{-3}$) and overfits at $h10$. A residual refiner that trains a neighbor-spillover ridge on the own-history GBM residual is statistically indistinguishable from the GBM at every horizon (DM $p = 0.07$, 0.69, 0.91, 0.70). Neither a feature-level nor a residual-level graph adds origin-time signal beyond own-history.

Takeaway. The graph null replicates on a second, structurally different market. The earnings lever is specific to the S&P 500 and does not generalize to HOSE under the available real announcement dates.

7 Limitations

Earnings dates. The earnings feature measures distance to the nearest scheduled release from an archive that records realized dates, not a point-in-time scheduling snapshot. A date-jitter check shows the gain does not depend on exact-date timing: perturbing every S&P 500 earnings date by a uniform random offset, the $h1$ QLIKE gain over own-history falls from 11.1% with true dates to 9.0% at plus or minus one trading day, 6.6% at plus or minus three days, and 4.5% at plus or minus five days, and stays positive at every horizon. A point-in-time scheduling calendar remains to be added to control rescheduling within the forecast window.

Vietnam coverage. The Vietnam earnings test draws on real announcement dates that are comprehensive only for 2025-01 through 2026-09, plus a sparse 2016-2026 news feed covering roughly 218 tickers. We therefore claim no incremental earnings value on HOSE under the available real dates, rather than a definitive full-history null.

Scope. The analysis is conditional on the Parkinson daily-OHLC regime and does not speak to intraday or option-implied signals.

8 Conclusion

We built a per-stock volatility forecasting model on a gamma-loss gradient-boosting machine over own-history features and asked where extra information helps. On the S&P 500, an External, forward-looking scheduled-earnings feature cuts QLIKE by 8.7 to 11.2% at every horizon (Diebold-Mariano $p < 0.001$), roughly seven times the size of any graph term, while a cross-firm graph feature adds nothing once own-history and the firm's own momentum are in the model. We replicated the study on 405 Vietnam HOSE tickers under an identical protocol: the graph null holds, and the earnings lever does not transfer, which localizes the earnings gain to the S&P 500 and shows that the graph carries no reliable signal on either market. The practical takeaway is that, for daily volatility, a forward-looking per-firm signal beats cross-firm wiring, and its value depends on the market's information and microstructure.

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